

Complete Course Notes

A beginner-friendly guide from Python fundamentals to practical projects.

LEARN. BUILD. INNOVATE.

Python Course Overview

Welcome to the Code Academy Python course. This course takes you from your first Python program through variables, data types, operators, collections, conditions, loops and functions, then combines those skills in practical projects.

Course outcome: By the end of the course, you should be able to read basic Python code, write your own programs, work with user input, make decisions, repeat tasks, organize code with functions, and build small practical applications.

How to use these notes

- Watch the lesson video first and use the notes as a reference.
- Type the examples yourself instead of only copying them.
- Run each example and observe the result.
- Complete the assignment after the lesson.
- Use the source-code PDF when you need to review the exact classroom examples.

Lesson 1: Introduction to Python and Your First Program

Lesson goal: Understand what Python is and write your first program using print().

Topics covered

- What Python is
- Writing and running Python code
- The print() function
- Displaying text
- Basic comments

Explanation

Python is a programming language used to give instructions to a computer. The print() function displays information in the output. Text is usually written inside quotation marks.

Key points to remember

- print() displays output.
- Strings are placed inside quotes.
- Python executes instructions in order.

Classroom example

```
<font color="#0369A1"><b>print</b></font>(<font color="#047857">"Hello World"</font>)
<font color="#0369A1"><b>print</b></font>(<font color="#047857">"Welcome to Code Academy"</font>)
<font color="#0369A1"><b>print</b></font>(<font color="#047857">"My name is Emmanuel Adedayo"</font>)
<font color="#0369A1"><b>print</b></font>(<font color="#047857">"Today we are learning Python"</font>)
<font color="#0369A1"><b>print</b></font>(<font color="#047857">"Starting with the first thing 'PRINT'"</font>)
```

Lesson 2: Variables and Data Types

Lesson goal: Learn how Python stores information and recognize common data types.

Topics covered

- Variables
- Strings
- Integers
- Floats
- Booleans
- Checking a data type with type()

Explanation

A variable is a named place used to hold information. Python can store different kinds of values, including text (str), whole numbers (int), decimal numbers (float), and True/False values (bool). The type() function lets you check a value's data type.

Key points to remember

- Variables have names and values.
- Common types include str, int, float and bool.
- Use type() to inspect a value's type.

Classroom example

```
<font color="#64748B"># STRING</font>
name = <font color="#047857">"Emmanuel"</font>
course = <font color="#047857">"python"</font>
platform = <font color="#047857">"code academy"</font>

<font color="#64748B"># INTEGER</font>
age = <font color="#C2410C"><b>20</b></font>
students = <font color="#C2410C"><b>500</b></font>

<font color="#64748B"># FLOAT</font>
price = <font color="#C2410C"><b>9.99</b></font>
height = <font color="#C2410C"><b>5.9</b></font>
percentage = <font color="#C2410C"><b>98.5</b></font>

<font color="#64748B"># BOOLEAN</font>
is_free = <font color="#7C3AED"><b>True</b></font>
is_paid = <font color="#7C3AED"><b>False</b></font>
is_active = <font color="#7C3AED"><b>True</b></font>

<font color="#64748B"># Print all of them</font>
<font color="#0369A1"><b>print</b></font>(name)
<font color="#0369A1"><b>print</b></font>(course)
<font color="#0369A1"><b>print</b></font>(platform)
<font color="#0369A1"><b>print</b></font>(age)
<font color="#0369A1"><b>print</b></font>(students)
<font color="#0369A1"><b>print</b></font>(price)
<font color="#0369A1"><b>print</b></font>(height)
<font color="#0369A1"><b>print</b></font>(percentage)
<font color="#0369A1"><b>print</b></font>(is_free)
```

```
<font color="#0369A1"><b>print</b></font>(is_paid)
<font color="#0369A1"><b>print</b></font>(is_active)

<font color="#64748B"># Check the datatype of any variable</font>
<font color="#0369A1"><b>print</b></font>(<font color="#0369A1"><b>type</b></font>(name))
<font color="#0369A1"><b>print</b></font>(<font color="#0369A1"><b>type</b></font>(course))
<font color="#0369A1"><b>print</b></font>(<font color="#0369A1"><b>type</b></font>(platform))
<font color="#0369A1"><b>print</b></font>(<font color="#0369A1"><b>type</b></font>(age))
<font color="#0369A1"><b>print</b></font>(<font color="#0369A1"><b>type</b></font>(students))
<font color="#0369A1"><b>print</b></font>(<font color="#0369A1"><b>type</b></font>(price))
<font color="#0369A1"><b>print</b></font>(<font color="#0369A1"><b>type</b></font>(height))
<font color="#0369A1"><b>print</b></font>(<font color="#0369A1"><b>type</b></font>(percentage))
<font color="#0369A1"><b>print</b></font>(<font color="#0369A1"><b>type</b></font>(is_free))
<font color="#0369A1"><b>print</b></font>(<font color="#0369A1"><b>type</b></font>(is_paid))
<font color="#0369A1"><b>print</b></font>(<font color="#0369A1"><b>type</b></font>(is_active))
```

Lesson 3: Operators, Input and a Simple Calculator

Lesson goal: Perform calculations, compare values, collect user input and build a basic calculator.

Topics covered

- Arithmetic operators
- Comparison operators
- input()
- Type conversion
- Building a simple calculator

Explanation

Operators allow a program to calculate and compare values. input() collects information from the user. Because input normally returns text, int() or float() can be used when a numeric value is required.

Key points to remember

- + can add numbers or join strings.
- Comparison operators produce True or False.
- Convert input to int or float when doing numeric calculations.

Classroom example

```
<font color="#64748B"># ARITHMETIC OPERATORS</font>
a = <font color="#C2410C"><b>20</b></font>
b = <font color="#C2410C"><b>6</b></font>

<font color="#0369A1"><b>print</b></font>(a + b)  <font color="#64748B"># ADDITION</font>
<font color="#0369A1"><b>print</b></font>(a - b)  <font color="#64748B"># SUBTRACTION</font>
<font color="#0369A1"><b>print</b></font>(a * b)  <font color="#64748B"># MULTIPLICATION</font>
<font color="#0369A1"><b>print</b></font>(a / b)  <font color="#64748B"># DIVISION</font>
<font color="#0369A1"><b>print</b></font>(a // b) <font color="#64748B"># FLOOR DIVISION</font>
<font color="#0369A1"><b>print</b></font>(a ** b) <font color="#64748B"># POWER</font>
<font color="#0369A1"><b>print</b></font>(a % b)  <font color="#64748B"># MODULUS</font>

<font color="#64748B"># COMPARISON OPERATORS</font>
<font color="#0369A1"><b>print</b></font>(a &gt; b)
<font color="#0369A1"><b>print</b></font>(a &lt; b)
<font color="#0369A1"><b>print</b></font>(a == b)
<font color="#0369A1"><b>print</b></font>(a != b)
<font color="#0369A1"><b>print</b></font>(a &gt;= b)
<font color="#0369A1"><b>print</b></font>(a &lt;= b)

<font color="#64748B"># INPUT</font>
name = <font color="#0369A1"><b>input</b></font>(<font color="#047857">"What is your name? "</font>)
course = <font color="#0369A1"><b>input</b></font>(<font color="#047857">"Which course do you like the most on this
age = <font color="#0369A1"><b>input</b></font>(<font color="#047857">"How old are you when you started this course?

<font color="#0369A1"><b>print</b></font>(<font color="#047857">"Hello "</font> + name + <font color="#047857">" You
      + course + <font color="#047857">" the most. You're one of the youngest, at "</font> + age + <font color="#047
```

```
<font color="#64748B"># BUILDING A SIMPLE CALCULATOR</font>
<font color="#0369A1"><b>print</b></font>(<font color="#047857">"\n--- SIMPLE CALCULATOR ---"</font>)
num1 = <font color="#0369A1"><b>float</b></font>(<font color="#0369A1"><b>input</b></font>(<font color="#047857">"En
num2 = <font color="#0369A1"><b>float</b></font>(<font color="#0369A1"><b>input</b></font>(<font color="#047857">"En

<font color="#0369A1"><b>print</b></font>(<font color="#047857">"Addition:"</font>, num1 + num2)
<font color="#0369A1"><b>print</b></font>(<font color="#047857">"Subtraction:"</font>, num1 - num2)
<font color="#0369A1"><b>print</b></font>(<font color="#047857">"Multiplication:"</font>, num1 * num2)
<font color="#0369A1"><b>print</b></font>(<font color="#047857">"Division:"</font>, num1 / num2)
```

Lesson 4: Lists and Tuples

Lesson goal: Store multiple values and access or update items in collections.

Topics covered

- Creating lists
- Indexing
- Negative indexing
- len()
- Updating list items
- Tuples

Explanation

A list stores multiple items in an ordered collection and can be changed. Indexing starts at 0, while negative indexes count from the end. A tuple is also ordered, but it is designed for values that should not be changed.

Key points to remember

- List indexes start at 0.
- Use negative indexes to count from the end.
- Lists can be updated; tuples are generally immutable.

Classroom example

```
<font color="#64748B"># CREATING LISTS</font>
students = [<font color="#047857">"Emmanuel"</font>, <font color="#047857">"Dave"</font>, <font color="#047857">"Joh
courses = [<font color="#047857">"HTML"</font>, <font color="#047857">"CSS"</font>, <font color="#047857">"JavaScrip
scores = [<font color="#C2410C"><b>85</b></font>, <font color="#C2410C"><b>92</b></font>, <font color="#C2410C"><b>7

<font color="#0369A1"><b>print</b></font>(students[<font color="#C2410C"><b>0</b></font>])
<font color="#0369A1"><b>print</b></font>(courses[<font color="#C2410C"><b>1</b></font>])
<font color="#0369A1"><b>print</b></font>(scores[-<font color="#C2410C"><b>1</b></font>])

<font color="#0369A1"><b>print</b></font>(<font color="#0369A1"><b>len</b></font>(students))

<font color="#64748B"># UPDATING AN ITEM</font>
students[<font color="#C2410C"><b>0</b></font>] = <font color="#047857">"Emmanuel Adedayo"</font>
<font color="#0369A1"><b>print</b></font>(students)

<font color="#64748B"># TUPLE</font>
days = (<font color="#047857">"Monday"</font>, <font color="#047857">"Tuesday"</font>, <font color="#047857">"Wednes
        <font color="#047857">"Friday"</font>, <font color="#047857">"Saturday"</font>, <font color="#047857">"Sunda
<font color="#0369A1"><b>print</b></font>(days)
```

Lesson 5: Dictionaries

Lesson goal: Store related information using key-value pairs and work with nested dictionaries.

Topics covered

- Creating dictionaries
- Keys and values
- Accessing values
- Checking whether a key exists
- Nested dictionaries
- items()

Explanation

A dictionary stores information as key-value pairs. Instead of using a numeric index, you use a key such as 'name' or 'email'. Dictionaries can also contain other dictionaries, which is useful for structured data.

Key points to remember

- Dictionaries use key-value pairs.
- Keys should be used to access the correct value.
- Nested dictionaries are useful for grouped records.

Classroom example

```
<font color="#64748B"># CREATING A DICTIONARY</font>
students = {
    <font color="#047857">"name"</font>: <font color="#047857">"Emmanuel Adedayo"</font>,
    <font color="#047857">"age"</font>: <font color="#C2410C"><b>20</b></font>,
    <font color="#047857">"course"</font>: <font color="#047857">"python"</font>,
    <font color="#047857">"email"</font>: <font color="#047857">"seatechnologiesmedia@gmail.com"</font>,
    <font color="#047857">"is_active"</font>: <font color="#7C3AED"><b>True</b></font>
}

<font color="#0369A1"><b>print</b></font>(students[<font color="#047857">"name"</font>])
<font color="#0369A1"><b>print</b></font>(students[<font color="#047857">"age"</font>])
<font color="#0369A1"><b>print</b></font>(students[<font color="#047857">"course"</font>])
<font color="#0369A1"><b>print</b></font>(students[<font color="#047857">"email"</font>])
<font color="#0369A1"><b>print</b></font>(students[<font color="#047857">"is_active"</font>])

<font color="#64748B"># CHECKING IF A KEY EXISTS</font>
<font color="#7C3AED"><b>if</b></font> <font color="#047857">"email"</font> <font color="#7C3AED"><b>in</b></font> s
    <font color="#0369A1"><b>print</b></font>(<font color="#047857">"email found "</font> + students[<font color="#0

<font color="#64748B"># DICTIONARY OF DICTIONARIES</font>
students = {
    <font color="#047857">"student1"</font>: {
        <font color="#047857">"name"</font>: <font color="#047857">"Emmanuel"</font>,
        <font color="#047857">"score"</font>: <font color="#C2410C"><b>85</b></font>,
        <font color="#047857">"grade"</font>: <font color="#047857">"Distinction"</font>
    },
    <font color="#047857">"student2"</font>: {
        <font color="#047857">"name"</font>: <font color="#047857">"Dave"</font>,
        <font color="#047857">"score"</font>: <font color="#C2410C"><b>72</b></font>,
```

```
        "grade": "Distinction"
    },
    "student3": {
        "name": "John",
        "score": 55,
        "grade": "Pass"
    }
}

print("\n--- ALL STUDENTS ---")
for student_id, info in students.items():
    print(info["name"] + " –
```

Lesson 6: Conditional Statements

Lesson goal: Make programs respond differently based on conditions.

Topics covered

- if
- else
- elif
- Comparison conditions
- User input
- Simple login logic

Explanation

Conditional statements allow a program to make decisions. if checks the first condition, elif checks another condition when needed, and else runs when none of the previous conditions is true.

Key points to remember

- if starts a decision.
- elif provides additional conditions.
- else handles the remaining case.

Classroom example

```
<font color="#64748B"># BASIC IF ELSE</font>
years = <font color="#C2410C"><b>18</b></font>

<font color="#7C3AED"><b>if</b></font> years &gt;= <font color="#C2410C"><b>18</b></font>:
    <font color="#0369A1"><b>print</b></font>(<font color="#047857">"Congratulations! You are an adult."</font>)
<font color="#7C3AED"><b>else</b></font>:
    <font color="#0369A1"><b>print</b></font>(<font color="#047857">"Sorry, you are not an adult yet."</font>)

<font color="#64748B"># IF ELIF ELSE</font>
score = <font color="#C2410C"><b>75</b></font>

<font color="#7C3AED"><b>if</b></font> score &gt;= <font color="#C2410C"><b>70</b></font>:
    <font color="#0369A1"><b>print</b></font>(<font color="#047857">"Distinction"</font>)
<font color="#7C3AED"><b>elif</b></font> score &gt;= <font color="#C2410C"><b>50</b></font>:
    <font color="#0369A1"><b>print</b></font>(<font color="#047857">"Pass"</font>)
<font color="#7C3AED"><b>elif</b></font> score &gt;= <font color="#C2410C"><b>40</b></font>:
    <font color="#0369A1"><b>print</b></font>(<font color="#047857">"Nearly Pass"</font>)
<font color="#7C3AED"><b>else</b></font>:
    <font color="#0369A1"><b>print</b></font>(<font color="#047857">"Fail"</font>)

<font color="#64748B"># WITH USER INPUT</font>
<font color="#0369A1"><b>print</b></font>(<font color="#047857">"\n--- GRADE CHECKER ---"</font>)
score = <font color="#0369A1"><b>float</b></font>(<font color="#0369A1"><b>input</b></font>(<font color="#047857">"E

<font color="#7C3AED"><b>if</b></font> score &gt;= <font color="#C2410C"><b>70</b></font>:
    <font color="#0369A1"><b>print</b></font>(<font color="#047857">"Excellent! You got a Distinction"</font>)
<font color="#7C3AED"><b>elif</b></font> score &gt;= <font color="#C2410C"><b>50</b></font>:
    <font color="#0369A1"><b>print</b></font>(<font color="#047857">"Good job! You Passed"</font>)
<font color="#7C3AED"><b>elif</b></font> score &gt;= <font color="#C2410C"><b>40</b></font>:
```

```
    <font color="#0369A1"><b>print</b></font>(<font color="#047857">"Almost there! Nearly Pass"</font>)
<font color="#7C3AED"><b>else</b></font>:
    <font color="#0369A1"><b>print</b></font>(<font color="#047857">"Do not give up! Keep studying"</font>)

<font color="#64748B"># LOGIN SYSTEM</font>
<font color="#0369A1"><b>print</b></font>(<font color="#047857">"\n--- LOGIN SYSTEM ---"</font>)
password = <font color="#0369A1"><b>input</b></font>(<font color="#047857">"Enter password: "</font>)

<font color="#7C3AED"><b>if</b></font> password == <font color="#047857">"python123"</font>:
    <font color="#0369A1"><b>print</b></font>(<font color="#047857">"Access granted! Welcome"</font>)
<font color="#7C3AED"><b>else</b></font>:
    <font color="#0369A1"><b>print</b></font>(<font color="#047857">"Wrong password. Access denied"</font>)
```

Lesson 7: Loops: for and while

Lesson goal: Repeat instructions efficiently and build repeated-output programs.

Topics covered

- for loops
- range()
- Looping through collections
- while loops
- Counters
- Multiplication tables

Explanation

Loops repeat code. A for loop is useful when you know what collection or range you want to go through. A while loop continues as long as its condition remains true. Always make sure a while loop can eventually stop.

Key points to remember

- for repeats over a sequence or range.
- while repeats while a condition is true.
- range(1, 11) produces values from 1 through 10.

Classroom example

```
<font color="#64748B"># FOR LOOP</font>
<font color="#7C3AED"><b>for</b></font> i <font color="#7C3AED"><b>in</b></font> <font color="#0369A1"><b>range</b><
    <font color="#0369A1"><b>print</b></font>(i)

<font color="#7C3AED"><b>for</b></font> i <font color="#7C3AED"><b>in</b></font> <font color="#0369A1"><b>range</b><
    <font color="#0369A1"><b>print</b></font>(<font color="#047857">"Number: "</font> + <font color="#0369A1"><b>str

courses = (<font color="#047857">"HTML"</font>, <font color="#047857">"CSS"</font>, <font color="#047857">"JAVASCRIP

<font color="#7C3AED"><b>for</b></font> course <font color="#7C3AED"><b>in</b></font> courses:
    <font color="#0369A1"><b>print</b></font>(<font color="#047857">"I am learning "</font> + course)

<font color="#64748B"># WHILE LOOP</font>
count = <font color="#C2410C"><b>1</b></font>

<font color="#7C3AED"><b>while</b></font> count &lt;= <font color="#C2410C"><b>100</b></font>:
    <font color="#0369A1"><b>print</b></font>(<font color="#047857">"Count is: "</font> + <font color="#0369A1"><b>s
    count = count + <font color="#C2410C"><b>1</b></font>

<font color="#64748B"># REAL WORLD – MULTIPLICATION TABLE</font>
<font color="#0369A1"><b>print</b></font>(<font color="#047857">"\n--- MULTIPLICATION TABLE ---"</font>)
number = <font color="#0369A1"><b>int</b></font>(<font color="#0369A1"><b>input</b></font>(<font color="#047857">"En

<font color="#7C3AED"><b>for</b></font> i <font color="#7C3AED"><b>in</b></font> <font color="#0369A1"><b>range</b><
    <font color="#0369A1"><b>print</b></font>(<font color="#0369A1"><b>str</b></font>(number) + <font color="#047857
```

Lesson 8: Functions

Lesson goal: Create reusable blocks of code with functions, parameters and return values.

Topics covered

- Defining functions
- Calling functions
- Parameters
- Arguments
- return
- Real-world functions

Explanation

A function is a reusable block of code. Parameters allow a function to receive information, while return sends a result back to the part of the program that called the function.

Key points to remember

- Functions reduce repeated code.
- Parameters receive values.
- return gives a result back.

Classroom example

```
<font color="#64748B"># BASIC FUNCTION</font>
<font color="#7C3AED"><b>def</b></font> greet():
    <font color="#0369A1"><b>print</b></font>(<font color="#047857">"Hello, welcome to Code Academy"</font>)

greet()
greet()
greet()

<font color="#64748B"># FUNCTION WITH PARAMETER</font>
<font color="#7C3AED"><b>def</b></font> greet_students(name):
    <font color="#0369A1"><b>print</b></font>(<font color="#047857">"Hello "</font> + name + <font color="#047857">"

greet_students(<font color="#047857">"Emmanuel"</font>)
greet_students(<font color="#047857">"Dave"</font>)
greet_students(<font color="#047857">"John"</font>)

<font color="#64748B"># REAL WORLD FUNCTION</font>
<font color="#7C3AED"><b>def</b></font> calculate_area(length, width):
    area = length * width
    <font color="#7C3AED"><b>return</b></font> area

room1 = calculate_area(<font color="#C2410C"><b>10</b></font>, <font color="#C2410C"><b>5</b></font>)
room2 = calculate_area(<font color="#C2410C"><b>8</b></font>, <font color="#C2410C"><b>6</b></font>)

<font color="#0369A1"><b>print</b></font>(<font color="#047857">"Room 1 area: "</font> + <font color="#0369A1"><b>st
<font color="#0369A1"><b>print</b></font>(<font color="#047857">"Room 2 area: "</font> + <font color="#0369A1"><b>st
```

Lesson 9: Working with Files and Building a Grade Calculator

Lesson goal: Combine functions, conditions, loops, lists and file handling in a practical project.

Topics covered

- Functions
- Lists
- Loops
- Conditions
- Input validation
- File writing
- A complete grade calculator

Explanation

This project combines several Python concepts. It collects student information, validates scores, calculates totals and averages, determines a grade, produces a remark, and saves the result to a text file.

Key points to remember

- Validate user input.
- Functions make the project easier to organize.
- open(..., 'a') appends new results to a file.

Classroom example

```
<font color="#64748B"># GRADE CALCULATOR</font>
<font color="#64748B"># Code Academy | SEA TECHNOLOGIES MEDIA</font>
<font color="#64748B"># Built with Python</font>

<font color="#7C3AED"><b>def</b></font> get_grade(average):
    <font color="#7C3AED"><b>if</b></font> average &gt;= <font color="#C2410C"><b>70</b></font>:
        <font color="#7C3AED"><b>return</b></font> <font color="#047857">"DISTINCTION"</font>
    <font color="#7C3AED"><b>elif</b></font> average &gt;= <font color="#C2410C"><b>50</b></font>:
        <font color="#7C3AED"><b>return</b></font> <font color="#047857">"PASS"</font>
    <font color="#7C3AED"><b>elif</b></font> average &gt;= <font color="#C2410C"><b>40</b></font>:
        <font color="#7C3AED"><b>return</b></font> <font color="#047857">"NEARLY PASS"</font>
    <font color="#7C3AED"><b>else</b></font>:
        <font color="#7C3AED"><b>return</b></font> <font color="#047857">"FAIL"</font>

<font color="#7C3AED"><b>def</b></font> get_remark(average):
    <font color="#7C3AED"><b>if</b></font> average &gt;= <font color="#C2410C"><b>70</b></font>:
        <font color="#7C3AED"><b>return</b></font> <font color="#047857">"Excellent performance! Keep it up!"</font>
    <font color="#7C3AED"><b>elif</b></font> average &gt;= <font color="#C2410C"><b>50</b></font>:
        <font color="#7C3AED"><b>return</b></font> <font color="#047857">"Good job! You can do even better!"</font>
    <font color="#7C3AED"><b>elif</b></font> average &gt;= <font color="#C2410C"><b>40</b></font>:
        <font color="#7C3AED"><b>return</b></font> <font color="#047857">"Almost there! Study harder next time!"</fo
    <font color="#7C3AED"><b>else</b></font>:
        <font color="#7C3AED"><b>return</b></font> <font color="#047857">"Do not give up! Keep pushing!"</font>

<font color="#7C3AED"><b>def</b></font> save_result(name, subjects, scores, average, grade):
```

```
    <font color="#7C3AED"><b>with</b></font> <font color="#0369A1"><b>open</b></font>(<font color="#047857">"results
        file.write(<font color="#047857">"="</font> * <font color="#C2410C"><b>40</b></font> + <font color="#047857"
        file.write(<font color="#047857">"Name: "</font> + name + <font color="#047857">"\n"</font>)

        <font color="#7C3AED"><b>for</b></font> i <font color="#7C3AED"><b>in</b></font> <font color="#0369A1"><b>ra
            file.write(subjects[i] + <font color="#047857">": "</font> + <font color="#0369A1"><b>str</b></font>(sco

        file.write(<font color="#047857">"Average: "</font> + <font color="#0369A1"><b>str</b></font>(<font color="#
        file.write(<font color="#047857">"Grade: "</font> + grade + <font color="#047857">"\n"</font>)
        file.write(<font color="#047857">"="</font> * <font color="#C2410C"><b>40</b></font> + <font color="#047857"

<font color="#7C3AED"><b>def</b></font> main():
    <font color="#0369A1"><b>print</b></font>(<font color="#047857">"="</font> * <font color="#C2410C"><b>45</b></fo
    <font color="#0369A1"><b>print</b></font>(<font color="#047857">"      CODE ACADEMY GRADE CALCULATOR"</font>)
    <font color="#0369A1"><b>print</b></font>(<font color="#047857">"      SEA TECHNOLOGIES MEDIA"</font>)
    <font color="#0369A1"><b>print</b></font>(<font color="#047857">"="</font> * <font color="#C2410C"><b>45</b></fo

    name = <font color="#0369A1"><b>input</b></font>(<font color="#047857">"\nEnter student full name: "</font>)

    subjects = [
        <font color="#047857">"Mathematics"</font>,
        <font color="#047857">"English Language"</font>,
        <font color="#047857">"Python Programming"</font>,
        <font color="#047857">"Web Development"</font>,
        <font color="#047857">"Problem Solving"</font>
    ]

    scores = []
    <font color="#0369A1"><b>print</b></font>(<font color="#047857">"\nEnter scores for each subject (0-100):\n"</fo

    <font color="#7C3AED"><b>for</b></font> subject <font color="#7C3AED"><b>in</b></font> subjects:
        <font color="#7C3AED"><b>while</b></font> <font color="#7C3AED"><b>True</b></font>:
            <font color="#7C3AED"><b>try</b></font>:
                score = <font color="#0369A1"><b>float</b></font>(<font color="#0369A1"><b>input</b></font>(subject

                <font color="#7C3AED"><b>if</b></font> <font color="#C2410C"><b>0</b></font> &lt;= score &lt;= <font
                    scores.append(score)
                    <font color="#7C3AED"><b>break</b></font>
                <font color="#7C3AED"><b>else</b></font>:
                    <font color="#0369A1"><b>print</b></font>(<font color="#047857">"Score must be between 0 and 100

            <font color="#7C3AED"><b>except</b></font> ValueError:
                <font color="#0369A1"><b>print</b></font>(<font color="#047857">"Please enter a valid number"</font>

    total = <font color="#0369A1"><b>sum</b></font>(scores)
    average = total / <font color="#0369A1"><b>len</b></font>(scores)
    grade = get_grade(average)
    remark = get_remark(average)

    <font color="#0369A1"><b>print</b></font>(<font color="#047857">"\n"</font> + <font color="#047857">"="</font> *
    <font color="#0369A1"><b>print</b></font>(<font color="#047857">"           RESULT FOR "</font> + name.upper())
    <font color="#0369A1"><b>print</b></font>(<font color="#047857">"="</font> * <font color="#C2410C"><b>45</b></fo

    <font color="#7C3AED"><b>for</b></font> i <font color="#7C3AED"><b>in</b></font> <font color="#0369A1"><b>range<
        <font color="#0369A1"><b>print</b></font>(f<font color="#047857">"{subjects[i]:&lt;25} {scores[i]}"</font>)

    <font color="#0369A1"><b>print</b></font>(<font color="#047857">"-"</font> * <font color="#C2410C"><b>45</b></fo
    <font color="#0369A1"><b>print</b></font>(f<font color="#047857">"{'Total:':&lt;25} {total}"</font>)
    <font color="#0369A1"><b>print</b></font>(f<font color="#047857">"{'Average:':&lt;25} {round(average, 2)}"</font
    <font color="#0369A1"><b>print</b></font>(f<font color="#047857">"{'Grade:':&lt;25} {grade}"</font>)
    <font color="#0369A1"><b>print</b></font>(<font color="#047857">"-"</font> * <font color="#C2410C"><b>45</b></fo
    <font color="#0369A1"><b>print</b></font>(f<font color="#047857">"Remark: {remark}"</font>)
    <font color="#0369A1"><b>print</b></font>(<font color="#047857">"="</font> * <font color="#C2410C"><b>45</b></fo

    save_result(name, subjects, scores, average, grade)
```

```
    <font color="#0369A1"><b>print</b></font>(<font color="#047857">"\nResult saved to results.txt successfully!"</f
    <font color="#0369A1"><b>print</b></font>(<font color="#047857">"\nThank you for using Code Academy Grade Calcul
    <font color="#0369A1"><b>print</b></font>(<font color="#047857">"Learn. Build. Innovate."</font>)

main()
```

Lesson 10: Final Project: Putting Python Skills Together

Lesson goal: Review the major concepts from the course and understand how they work together in a real program.

Topics covered

- Variables
- Data types
- Operators
- Collections
- Conditions
- Loops
- Functions
- Input
- File handling
- Project thinking

Explanation

The final project review shows that programming concepts are not isolated. Variables, collections, functions, conditions and loops can be combined to solve a real problem. This is the mindset used when building larger applications.

Key points to remember

- Break a problem into smaller parts.
- Combine concepts gradually.
- A project is where individual skills become useful together.

Classroom example

```
<font color="#64748B"># FINAL PROJECT REVIEW</font>
<font color="#64748B"># The Grade Calculator combines the major Python concepts taught</font>
<font color="#64748B"># throughout the course.</font>

<font color="#64748B"># 1. Variables store information.</font>
student_name = <font color="#047857">"Emmanuel"</font>

<font color="#64748B"># 2. Lists store multiple values.</font>
subjects = [<font color="#047857">"Python"</font>, <font color="#047857">"HTML"</font>, <font color="#047857">"Probl

<font color="#64748B"># 3. Functions organize reusable logic.</font>
<font color="#7C3AED"><b>def</b></font> calculate_average(scores):
    <font color="#7C3AED"><b>return</b></font> <font color="#0369A1"><b>sum</b></font>(scores) / <font color="#0369A

<font color="#64748B"># 4. Conditions make decisions.</font>
<font color="#7C3AED"><b>def</b></font> get_status(average):
    <font color="#7C3AED"><b>if</b></font> average &gt;= <font color="#C2410C"><b>70</b></font>:
        <font color="#7C3AED"><b>return</b></font> <font color="#047857">"Excellent"</font>
```

```
    elif average >= 50:
        return "Pass"
    else:
        return "Needs Improvement"

# 5. Use the functions.
scores = [85, 78, 9
average = calculate_average(scores)
status = get_status(average)

print("Student:", student_name)
print("Subjects:", subjects)
print("Average:", round
print("Status:", status)

# This is how individual Python concepts become a complete program.
```

Two Course Projects

Project 1 — Simple Calculator: accepts two numbers from the user and performs addition, subtraction, multiplication and division. It reinforces input, type conversion, operators and output.

Project 2 — Grade Calculator: collects a student's scores, calculates the total and average, determines a grade and remark, displays the result, and saves it to a text file. It reinforces conditions, loops, functions, lists, validation and file handling.

Remember: Do not only memorize Python syntax. Understand what each part of the program is doing, then practice changing the examples to solve your own problems.